# MTH304- Lecture 01.

## 01. Introduction to Statics.

* **Mechanics:-**

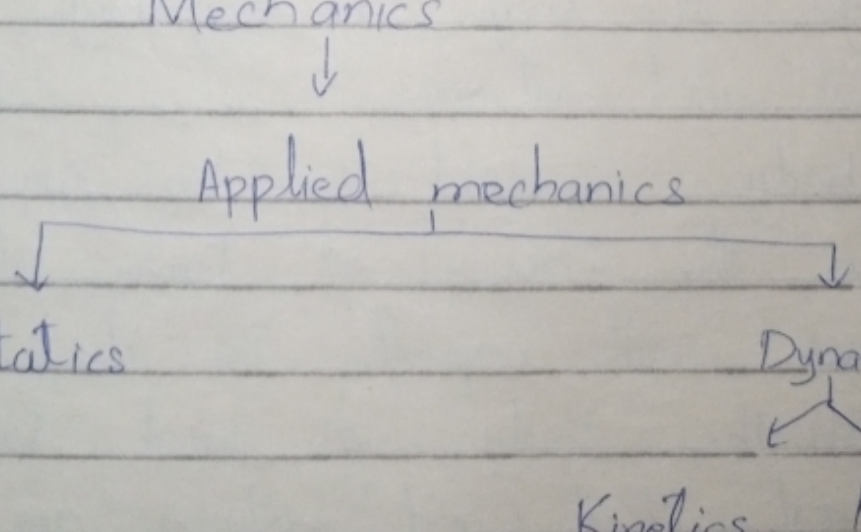
Mechanics is the branch of physical Sciences which studies the action of force on bodies. Under the action of these forces, bodies are in state of rest or of motion relative to some frame of reference.

Mechanics is further divided into two areas.

1) Statics 2) ~~Statics~~ Dynamics.

* **Statics:-** It is the branch of mechanics which deals with forces acting on a body which is at rest.

* **Dynamics:-** It is that branch of mechanics which deals with accelerated motion.



## 02. Fundamental concepts of statics:-

Some basic definitions:-

1) Body:-

Matter which supplies limited portion of space is called a body.

2) Mass:-

The quantity of a matter in a body is called its mass.

3) Particle:-

A body of negligibly small dimension is called particle.

4) Rigid body:-

A body is said to a rigid body if the distance between any two particles of the body remains unchanged so that the body does not expand or contract or change its shape in any way.

5). Rest and motion:-

A body is said to be in motion if, in the course of time, its position changes with respect to its surrounding otherwise the body is said to be in rest. Rest and motion are relative

concepts and are meaningful only when referred to surrounding bodies.

03. Inertial and non-inertial frame of reference.

Recall: Newton's First Law of Motion:-
An object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.
Also known as Law of Inertia.

Examples:-
You know that a person sitting on a bus watching a ball roll across the ground will have a very different ~~velocity~~ measurement of the ball's velocity than a person standing outside the bus would measure the ball's velocity.
Then the person inside the bus and the person outside the bus are measuring from different frames of reference.

* The laws of physics seems to momentarily break down for you sitting on the bus. In reality, what has happened

is that your frame of reference has been compromised.

1. Inertial frame of reference:-

An inertial frame of reference is a frame of reference in which the law of inertia and other physics laws are valid. Any frame moving at a constant velocity relative to another frame is also an inertial frame of reference

Example:-

When the breaks are applied to the bus, the bus undergoes a negative acceleration. At this moment, it becomes a non-inertial frame of reference.

2. Non-inertial reference frame.-

It is a reference frame in which the law of inertia does not hold.

* Although the bus accelerates towards the front of bus, there is no net force causing the acceleration.
* But if you are sitting on the bus, you observe the ball accelerating forward. That would imply to you as you sit

on the bus that there is a net force forward on the ball.

* The reason there appears to be a net force on the ball is that you are observing the motion of ball in the non-inertial reference frame.

# MTH304 - Lecture # 2

01. Basic concepts of force.

What is a force?

A force is a pull or push acting on an object that changes the motion of the object.

* Types of forces:-

1). Contact force:- Forces that act through direct contact between two objects.

ie Applied forces, Forces.

2) Long range forces:-

Forces that can act over distances.

ie. Gravity, Electromagnetic Force (EMF)

Newton's 2nd Law of Motion.

* Newton discovered the idea of a force
* He found the force is proportional to acceleration of an object (more Force = more Acceleration)
* He found the force is proportional to the mass of the object (more mass = more force needed.)
* Moving objects accelerate when an unbalanced force (F) acts on them. The stronger the force, the greater the acceleration (a)

1.

Also, the greater the mass (m) the greater the force required to change the motion.

Force = Mass x Acceleration

$F = ma$

* Force is a vector quantity and therefore has magnitude and direction.
* The direction of the force is the same as the direction of the acceleration

S.I. Unit of force.

* The unit of force is Newton (N).

$1N = 1kgm/s^{+2}$

* A Newton (N) is defined as the amount of force required to accelerate 1kg of mass at a rate of $1 ms^{-2}$.
* When multiple forces are acting on an object. The Net force is the amount of force that is left after adding all the forces on the object

Net Force ($F_{NET}$) = Resultant Force ($F_R$)

Balanced Forces.

Balanced forces are the forces that are equal and opposite so

that they cancel out.

10N East → 10N West ←

Net force = +10N + (-10N) = 0

## UNBALANCED FORCES.

Unbalanced forces are the forces that when they do not cancel out and cause a change in the motion of objec

30N East → 10N West ←

$F_{NET}$ = 30N + (-10N) = 20N East.

## INERTIA

Inertia measures the tendency of an object to resist changes in motion.

* Galileo came up with the idea of Inertia.
* Objects don not want their motion to change.
* Mass measures how much inertia an object has. (More mass = More inertia)

### Newton's First law of Motion. (Law of Inertia)

* If no unbalanced forces act on a

moving object, then the object will continue to move with a constant velocity (constant speed in a straight line). If an object is at rest it will stay at rest.

* Newton took his concept of forces and combined it with Galileo's idea of inertia.

## EQUILIBRIUM.

* First condition for equilibrium.

Example:-

If the net force acting on the object is zero.

$F_{NET} = 0$ , $a = 0$

The object is either stationary ($v = 0$) or travelling with a constant velocity ($v$ = constant).

## Mass and Weight

Mass is the amount of matter that an object possess. Mass does not change with location.

Weight is the gravitational force that a large body (such as a planet) exerts

on another object.
Weight is a Force. It is also measured in Newtons.
Weight = Mass × Accleration due to gravity.

Gravity (Newton's 2nd Law)
$W = mg$
Weight does change with location. ("g" will change with location).

02. Equivalent Force Systems:-

1) Point of application of a Force:-
The point of the body where the force acts is called the point of application.

2) Line of action of a force:-
The line passing through the point of application in the direction of force is called its line of action.

3) Force system:- A set of forces acting on a particle or rigid body is called a force system or system of forces.

4) Equivalent force system:- Two force system are said to be equivalent if one system can be replaced by the

other without disturbing state of rest or motion of a body.

3) Law of Superposition and Transmissibility

Law of Superposition:-

The action of a given system on a body remains unchanged if we add to or subtract from them another system of forces in equilibrium.

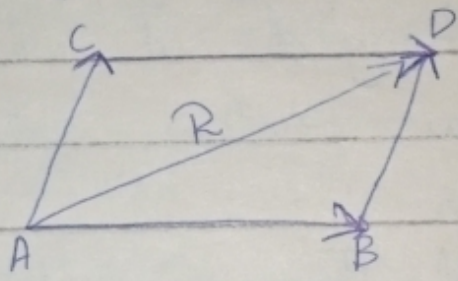
Principle of Transmissibility of Forces:-

The point of application of a force, acting on a rigid body can be transfe to any other point of the body lying on the line of action of the force without changing the effect.

# MTH304 - Lecture #3

## 01) Parallelogram Law:-

If two forces act on a body at a pt A along the adjacent sides of a parallelogram, then their resultant is the diagonal of the parallelogram with A as initial point.




$$\vec{R} = \overrightarrow{AB} + \overrightarrow{BD}$$

$$\vec{R} = \overrightarrow{AC} + \overrightarrow{BD}$$

## 2) Explanation of Parallelogram Law:-

### Vector:-

The term vector is used by scientists to indicate a quantity (such as displacement or velocity or force) that has both magnitude and direction.

### Representation of Vectors-

A vector is often represted by an arrow or a directed line segment. The length of the arrow represents the magnitude of the vector.

2) The arrow points in the direction of the vector.

Denoting a Vector:

We denote a vector by either:

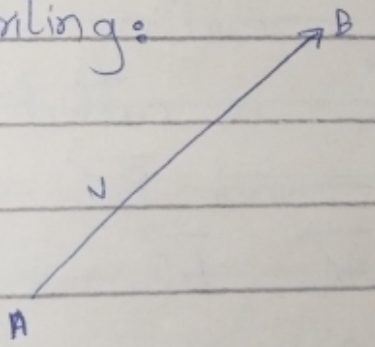
* Printing a letter in boldface (v).
* Putting an arrow above the letter ($\vec{v}$).

For instance, a suppose a particle moves along a line segment ~~from~~ along point A to point B.

The corresponding displacement vector $\vec{v}$ has initial point A (the tail) and a terminal point B (the tip).

We indicate this by writing:

$$v = \overrightarrow{AB}$$




Example:-

Notice the vector $u = \vec{CD}$ has the same length and the same direction as v even though it is in a different posit

* We say v and u are equivalent or equal and write $u = v$.

Zero vector:-

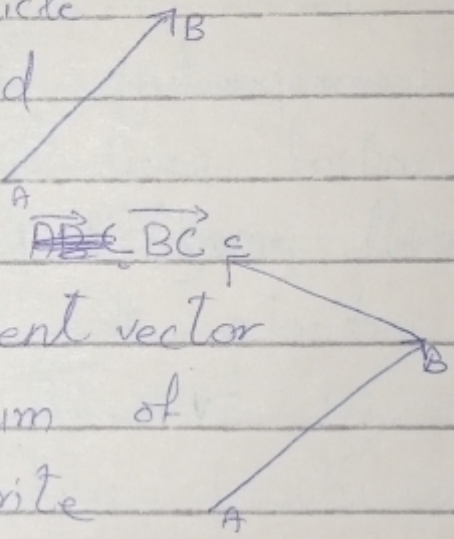
The zero vector, denoted by 0, has length 0.

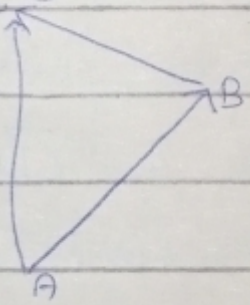
* It is the only vector with no specific direction.

## COMBINING VECTORS.

Suppose a particle moves from A to B. So, its displacement vector is $\overrightarrow{AB}$. Then, the particle changes direction and moves from B to C with displacement vector $\overrightarrow{BC}$. The resulting displacement vector $\overrightarrow{AC}$ is called the sum of $\overrightarrow{AB}$ and $\overrightarrow{BC}$. We write

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$



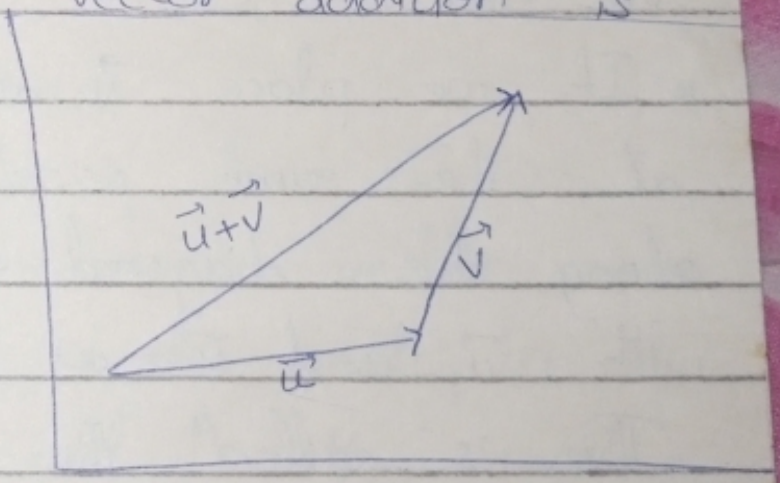

## ADDING VECTORS:-

In general, if we start with vectors

u and v, we first move v, so that its tail coincides with the tip of u. and define the sum of u and v as follows:

If $\vec{u}$ and $\vec{v}$ are vectors positioned so the initial point of $\vec{v}$ is at the terminal point of $\vec{u}$, then the sum $\vec{u}+\vec{v}$ is the vector from the initial point of u to the terminal point of v.

The definition of vector addition is illustrated here.




Triangle Law:-

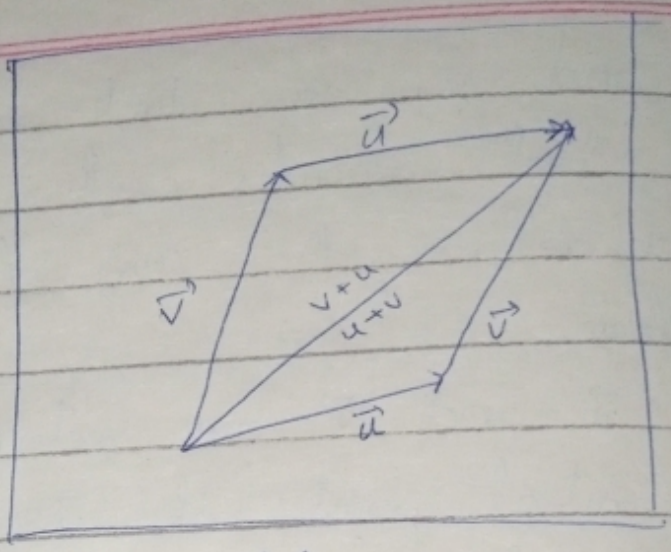
This definition is also sometimes called the Triangle law.

VECTOR ADDITION.

Here, we start with the same vectors $\vec{u}$ and $\vec{v}$ as earlier and draw a copy of $\vec{v}$ with the same initial point

16.

as $\vec{u}$.




Completing the parallelogram, we see that.

$$\vec{u} + \vec{v} = \vec{v} + \vec{u}$$

This also gives another way to construct the sum:

* If we place $\vec{u}$ and $\vec{v}$ so they start at the same point, then $\vec{u}+\vec{v}$ lies along the diagonal of the parallelogram with $\vec{u}$ and $\vec{v}$ as sides.

This is called the Parallelogram law.

## MULTIPLYING VECTORS.

It is possible to multiply a vector by a real number c. In this context, we call this real number c as a scalar - to distinguish it from vector.

Example:-

We want $2\vec{v}$ to be the same

17.

vector as $\vec{v} + \vec{v}$, which has the same direction as $\vec{v}$ but is twice as long. In general, we multiply a vector by a scalar as follows:

## SCALAR MULTIPLICATION.

If $c$ is a scalar and $\vec{v}$ is a vector the scalar multiple $c\vec{v}$ is:

The vector whose length is $|c|$ times the length of $\vec{v}$ and whose direction is the same as $\vec{v}$ if $c > 0$ and is opposite to $\vec{v}$ if $c < 0$.

If $c = 0$ or $\vec{v} = 0$ then $c\vec{v} = 0$

Notice that two non-zero vectors are parallel if they are scalar multiples of one another. We call it the negative of $\vec{v}$.

## SUBTRACTING the VECTORS.

By the difference $\vec{u} - \vec{v}$ of two vectors we mean:

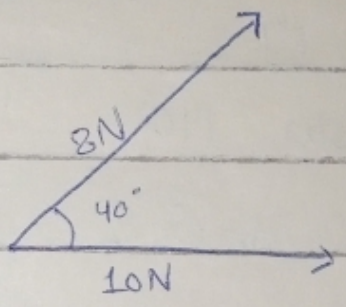
$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$$

Alternatively, since $\vec{v} + (\vec{u} - \vec{v}) = \vec{u}$, the vector $\vec{u} - \vec{v}$, when added to $\vec{v}$ gives $\vec{u}$.

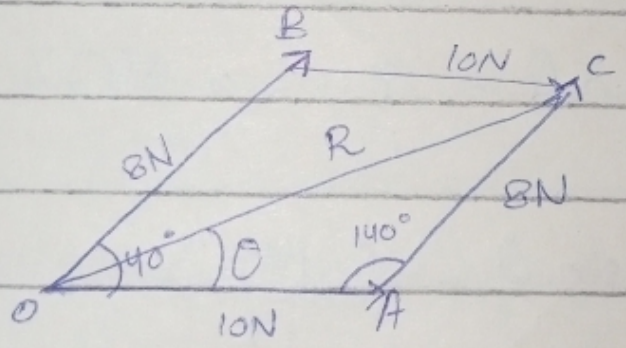
# MTH304 - Lecture # 4

## 01. Parallelogram Law Example 1.

Two forces of magnitude 8N and 10N are acting on a particles as shown in the figure. Find the resultant forces by applying parallelogram law. Find also the angle made by the resultant with 10N force.




**Solution:-**




In parallelogram, the sum of adjacent angles is 180°

Here the given angle is 40°. So the other will be 140°. i.e. 180° − 40° = 140°

$\Delta OAC$

Law of cosines $\rightarrow \vec{OA}^2 + \vec{OB}^2 - 2(\vec{OA})(\vec{OB})\cos\theta$

$R^2 = 8^2 + 10^2 - 2(8)(10)(\cos 140^\circ)$

$= 64 + 100 - 160(\cos 140^\circ)$

$= 164 - 160(\cos 140°)$

$= 164 + 122.567$

$R^2 = 286.57$

$\boxed{R = 17N}$

$\theta = ?$

Using law of sines

$$\frac{R}{\sin\theta} = \frac{\overrightarrow{OC}}{\sin\theta}$$

$$\frac{17}{\sin 140°} = \frac{8}{\sin\theta}$$

$$\sin\theta = \frac{8}{17} \times \sin 140°$$

$$\sin\theta = \frac{8}{17} \times 0.6427$$

$$\sin\theta = \frac{5.1423}{17}$$

$$\sin\theta = 0.302$$

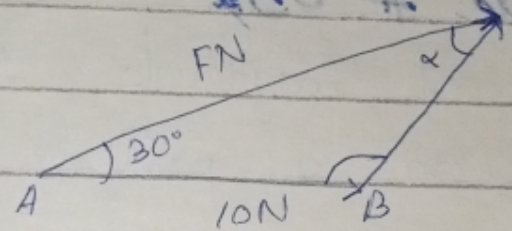
$$\theta = \sin^{-1}(0.3)$$

$$\boxed{\theta = 17.4°}$$

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## Example #2,

Two forces of magnitude 10N and 15N are the resolved parts of force of magnitude FN as shown in the figure. Find the magnitude of Force F.




## Solution:-

Using law of sines

$$\frac{15}{\sin 30^\circ} = \frac{10}{\sin \alpha}$$

$$\sin \alpha = \frac{10}{15} \times \sin 30 = \frac{1}{3}$$

$$\alpha = \sin^{-1}\left(\frac{1}{3}\right)$$

$$\boxed{\alpha = 19.5^\circ}$$

$$\beta = 180^\circ - 30^\circ - 19.5^\circ$$

$$\beta = 130.5^\circ$$

Now using law of cosines

$F^2 = 10^2 + 15^2 - 2 \times 10 \times 15 \times \cos 130.5°$

$F^2 = 100 + 125 - 300 (\cos 130.5°)$

$= 225 + 194.83$

$F^2 = 419.83$

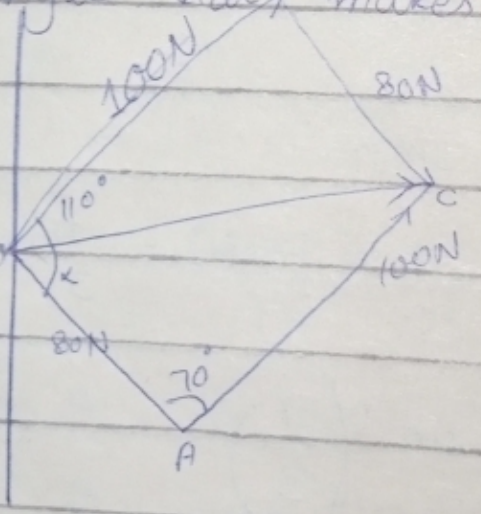
$\boxed{F = 20.4}$

## Example # 03

Two forces of magnitude 100N and 80N are acting on a particle as shown in the figure.

i) Find the magnitude by applying the parallelogram law.

ii) Find the angle that R makes with vertical




## Solution:-

→ 70° was given.

Using law of cosines.

$R^2 = 100^2 + 80^2 - 2(100)(80) \cos 70°$

$R^2 = \sqrt{16400 - 16000(\cos 70°)}$

$R = 104.54N$

Using law of sines

$$\frac{\sin\alpha}{100} = \frac{\sin 70^\circ}{104.54}$$

$$\sin\alpha = \frac{0.94}{104.54} \times 100$$

$$\sin\alpha = \frac{94}{104.54}$$

$$\alpha = \sin^{-1}(0.90)$$

$$\boxed{\alpha = 64^\circ}$$

## The sine and Cosine rule.

### Introduction :-

To solve a triangle is to find all its lengths and all its angles. The **sine rule** is used when we are given either

a) two angles and one side.

OR b) Two sides and a non-included angle.

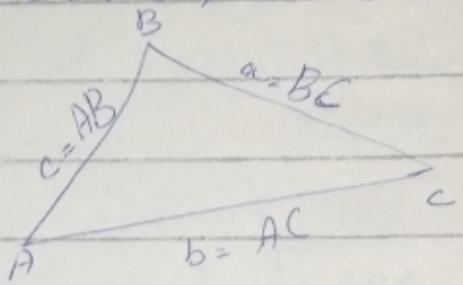
The **cosine rule** is used when we are given either

a) Three sides

b) two sides and the included angle.

## 1. The sine rule:-

Study the triangle shown below. Let B stands for the angle at B. Let C stand for angle at C and so on. Also let $b = AC$, $a = BC$, $c = AB$.




The sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

## Example:-

In triangle ABC, $B = 21°$, $C = 46°$, $AB = 9$cm. Solve this triangle.

## Solution:-

We are given two angles and one side and so the sine rule can be used. Furthermore, since the angles in any triangle must add up to 180° then angle A must be 113°. We know that $c = AB = 9$. Using the sine rule.

$$\frac{a}{\sin 113^\circ} = \frac{b}{\sin 21^\circ} = \frac{9}{\sin 46^\circ}$$

$$b = \sin 21^\circ \times \frac{9}{\sin 46^\circ} = 4.484\text{ cm}$$

$$a = \sin 113^\circ \times \frac{9}{\sin 46^\circ} = 11.517\text{ cm}$$

## 2. The Cosine rule:-

Using the triangle shown above:

The cosine rule:-

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

**Example:-**

In triangle ABC, AB = 42cm, BC = 37cm and AC, 26cm. Solve this triangle.

**Solution:-**

We are given three sides of the triangle and so the cosine rule can be used.

Writing $a = 37$, $b = 26$, $c = 42$ we have

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$37^2 = 26^2 + 42^2 - 2(26)(42)\cos A$$

$$\cos A = \frac{10.71}{2184} = 0.4904$$

$$A = 60.63^\circ$$